Git, GitHub, and Codespaces

What we will build—and what you will be able to do

The finished repository:

```
growth-calculator/
  README.md
  growth.py
  test_growth.py
  .gitignore
```

By the end, you can

- create a repository and a Codespace;
- edit, run, and test Python code;
- inspect, stage, and commit changes;
- push work to GitHub and pull new work;
- recognize the state of a Git repository.

We will learn Git by repeatedly using it on one small project.

Why not save a new file every time?

```
growth.py
growth_final.py
growth_final_v2.py
growth_final_v2_fixed.py
growth_REALLY_FINAL.py
```

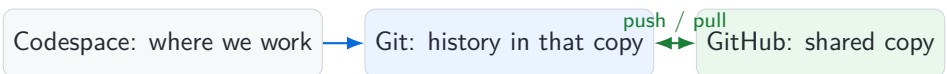
This does not tell us:

- what changed between versions;
- why a change was made;
- which code generated a result;
- whether two collaborators changed the same file.

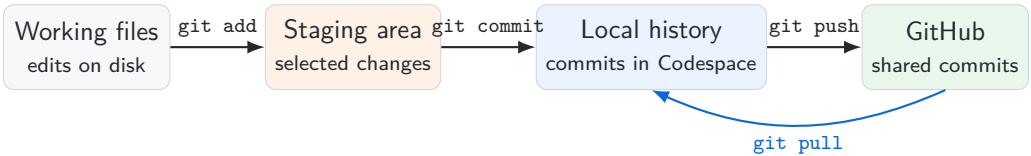
Git stores named snapshots of the project and a history connecting them.

Three tools, three different jobs

- Git records versions of files and connects them in a history.
- GitHub hosts a repository online so it can be shared and synchronized.
- Codespaces provides a cloud computer and VS Code editor in the browser.



The Git pipeline: where is a change?



Saving a file, committing it, and pushing it are three separate actions. Keeping them apart prevents most of the confusion that follows.

Step 1: open GitHub's repository form

In the GitHub website:

1. Sign in at `github.com`.
2. In the upper-right corner, click `+`.
3. Select **New repository**.

The menu that opens lists:

```
New repository
Import repository
New codespace
New gist
```

A repository is the project folder plus the Git history attached to it.

Step 2: configure and create the repository

Fill in the form as follows.

Owner	`your-username`
Repository name	`growth-calculator`
Description	`A Python growth calculator`
Visibility	Private
Add a README file	yes

Then click **Create repository**.

Checkpoint: the repository exists on GitHub.

We add a README because

- it explains the project to a visitor;
- it gives the repository a first file;
- GitHub creates the first commit for us.

Step 3: start a Codespace

From the repository page:

1. Click the green **Code** button.
2. Select the **Codespaces** tab.
3. Click **Create codespace on main**.

GitHub opens a new browser tab. Starting the cloud computer can take a minute.

The Codespace contains a working copy of the GitHub repository.

Orient yourself inside Codespaces

We will use three areas:

- the Explorer, to create and open files;
- the editor, to write code;
- the terminal, to run commands.

Open the terminal with **Terminal** → **New Terminal**.

Inspect the starting repository

Type in the terminal:

```
$ pwd
/workspaces/growth-calculator
$ ls
README.md
$ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

Checkpoint: both copies are synchronized.

Read the output as follows:

- `pwd`: show the current folder.
- `ls`: list the files in it.
- `main`: the current branch.
- `origin/main`: GitHub's corresponding branch.
- `clean`: no unsaved Git changes.

Read the existing history

```
$ git log --oneline
a1b2c3d (HEAD -> main, origin/main)
        Initial commit
```

The output says that

- `a1b2c3d` is the shortened commit identifier;
- `HEAD -> main` is where our Codespace currently points;
- `origin/main` is where GitHub currently points;
- `Initial commit` is the saved description.

Both labels point to the same commit, so the two copies are synchronized.

Create the program: `growth.py`

In the Explorer:

1. Click the **New File** icon.
2. Name the file `growth.py`.
3. Enter the code shown here.
4. Save with **Ctrl+S**.

No package installation is needed.

```python
def percentage_change(old_value, new_value):
    """Return percentage change from old to new."""
    if old_value == 0:
        raise ValueError("old_value cannot be zero")

    return 100 * (new_value - old_value) / old_value


if __name__ == "__main__":
    result = percentage_change(100, 105)
    print(f"Percentage change: {result}%")
```

Run the program before saving a Git version

```
$ python growth.py
Percentage change: 5.0%
```

Try a different pair of values by changing:

```
result = percentage_change(100, 105)
```

Python executed the file and the program produced the expected result. Git has not saved a new version, and GitHub has not received anything.

Running code and recording its version are separate actions.

Ask Git what changed

```
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what
   will be committed)
        growth.py

nothing added to commit
```

"Untracked" means that the file exists in the Codespace, that Git sees it, and that it has never been included in a commit.

Note that `git diff` shows changes to tracked files. It will not display the contents of this new, untracked file.

Stage the file, then review exactly what is staged

```
$ git add growth.py
$ git status
Changes to be committed:
        new file:   growth.py

$ git diff --staged
diff --git a/growth.py b/growth.py
new file mode 100644
--- /dev/null
+++ b/growth.py
@@ ...
+def percentage_change(old_value, new_value):
```

`git add` selected `growth.py` for the next commit, but it did not create the commit. `git diff --staged` previews the exact snapshot.

State: staged, not committed.

Commit: save one coherent change

```
$ git commit -m "Add growth calculator"
[main d4e5f6a] Add growth calculator
 1 file changed, 11 insertions(+)
 create mode 100644 growth.py

$ git log --oneline
d4e5f6a (HEAD -> main) Add growth calculator
a1b2c3d (origin/main) Initial commit
```

The commit identifier is `d4e5f6a`. `HEAD -> main` moved to the new commit; `origin/main` did not move. The commit therefore exists locally, but not yet on GitHub.

State: committed locally.

Push: send local commits to GitHub

```
$ git push
Enumerating objects: 4, done.
Writing objects: 100% (3/3), done.
To https://github.com/your-username/
growth-calculator.git
   a1b2c3d..d4e5f6a  main -> main
```

Then return to the GitHub tab and refresh the repository page.

Verify on GitHub that

1. `growth.py` is visible;
2. the commit message is visible;
3. the history shows two commits.

A successful local commit does not imply a successful push.

Checkpoint: Codespace and GitHub contain the new commit.

Create package-free tests: `test_growth.py`

Create a new file named `test_growth.py`.

Each `assert` states a result that must be true. If one is false, Python stops with an error.

`growth` refers to our own `growth.py` file. The tests use only Python itself.

```python
from growth import percentage_change


assert percentage_change(100, 105) == 5
assert percentage_change(100, 90) == -10
assert percentage_change(100, 100) == 0

print("All tests passed.")
```

Run the tests—and see what failure looks like

When all expectations are correct:

```
$ python test_growth.py
All tests passed.
```

Temporarily change the first expected result from 5 to 6 and rerun.

A failing test:

```
$ python test_growth.py
Traceback (most recent call last):
  File "test_growth.py", line 4
    assert percentage_change(100,105) == 6
AssertionError
```

Restore 5 and rerun until all tests pass.

A test converts an expected result into a check the computer can repeat.

Record the tested change

```
$ git status
$ git diff
$ python test_growth.py
All tests passed.
$ git add test_growth.py
$ git diff --staged
$ git commit -m "Add tests for growth calculator"
$ git push
```

The order matters:

1. identify changed files;
2. inspect their contents;
3. verify the code works;
4. select the intended change;
5. review, commit, and share it.

A good commit is small, coherent, reviewed, and tested.

Simulate a collaborator by editing on GitHub

On the GitHub repository page:

1. Click `README.md`.
2. Click the pencil icon **Edit this file**.
3. Add the text shown at right.
4. Click **Commit changes...**.
5. Enter `Describe the project` and confirm.

The new content of `README.md`:

```
# Growth calculator

A small Python program that
calculates percentage changes.
```

GitHub is now one commit ahead of the Codespace.

Fetch, inspect, then pull the new commit

```
$ git fetch
From https://github.com/.../growth-calculator
   6141a87..5f89e4d  main -> origin/main

$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit,
and can be fast-forwarded.
  (use "git pull" to update your local branch)

$ git pull
Updating 6141a87..5f89e4d
Fast-forward
 README.md | 3 +++
 1 file changed, 3 insertions(+)
```

1. `git fetch` contacts GitHub and refreshes `origin/main`; it does not change your files.
2. `git status` now sees that GitHub is one commit ahead.
3. `git pull` moves `main` forward and updates `README.md`.

"Fast-forward" means there was no competing local commit to merge.

What is `.gitignore`?

`.gitignore` is a plain-text file that tells Git which files or folders it should not track. Each line names a path or pattern to ignore; that line is sometimes called a "rule."

For this project, the complete file is one line:

```
__pycache__/
```

It means: "Git, ignore the `__pycache__/` directory that Python created."

Nothing is deleted. The directory stays on the computer; Git simply stops offering to include it in commits.

Create, check, and commit `.gitignore`

Create the file:

```
$ printf "__pycache__/\n" > .gitignore
$ cat .gitignore
__pycache__/
```

`printf` writes the line into `.gitignore`; `cat` checks what was written.

Check the result:

```
$ git status --short
?? .gitignore
```

The cache is no longer listed because Git is ignoring it.

Share the instruction:

```
$ git add .gitignore
$ git commit -m "Ignore Python cache"
$ git push
```

We commit `.gitignore` so everyone using the repository gets the same instruction: `.gitignore` is tracked, `__pycache__/` is ignored.

If a push is rejected, synchronize first

```
$ git push
! [rejected] main -> main (fetch first)
error: failed to push some refs
```

This means GitHub has commits that your Codespace does not have. Git refuses to overwrite them.

The safe response is

```
$ git status
$ git pull
# Read any message from Git carefully.
$ python test_growth.py
$ git push
```

If Git reports a conflict, stop and resolve the marked files before pushing.

Do not solve this with force: for this course workflow, never use `git push -force`, which can discard shared history.

What is a merge conflict?

A conflict occurs when Git cannot decide how to combine competing edits automatically. Git marks the disputed region:

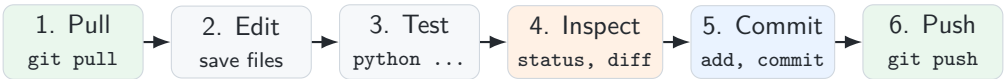
```
<<<<< HEAD
Percentage change calculator
=======
Growth rate calculator
>>>>> origin/main
```

To resolve it,

1. read both versions;
2. edit the file to the intended final content;
3. remove all conflict markers;
4. run the program and tests;
5. stage and commit the resolution.

A conflict is a request for human judgment, not evidence that work was lost.

The daily workflow



Before the commit, ask

- does the program run?
- do the tests pass?
- did I inspect every changed line?

After the commit, ask

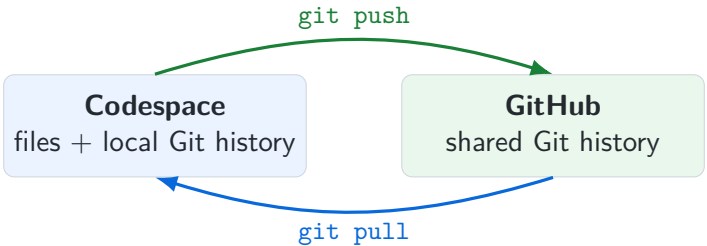
- is the message meaningful?
- did the push succeed?
- can I see the commit on GitHub?

Command guide: question first, command second

Question	Command
What is the repository's current state?	`git status`
What unstaged tracked lines changed?	`git diff`
What exactly is selected for the next commit?	`git diff –staged`
Which changes should enter the next commit?	`git add <file>`
How do I save the staged snapshot?	`git commit -m "message"`
What commits exist in the history?	`git log –oneline`
How do I receive GitHub's newer commits?	`git pull`
How do I send my local commits to GitHub?	`git push`

When uncertain, start with `git status`; read Git's message before acting.

The one picture to remember



Pull → Edit → Test → Inspect → Commit → Push

Use it on every project until the workflow becomes routine.